



Brain Tumor MRI Detection

Using Classical Computer Vision and Transfer Learning

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Abstract

PROJECT TITLE:

Brain Tumor MRI Detection Using Classical Computer Vision and Transfer Learning

PURPOSE:

This project investigates two automated approaches to detecting brain tumors in MRI scans:

- 1.classical computer vision (thresholding, morphology, segmentation, handcrafted features + logistic regression), and*
- 2.a deep learning approach using ResNet-18 transfer learning.*

MISSION:

Empowering small businesses to rise above the competition through bold and authentic digital marketing strategies.

GOAL:

Create a hybrid pipeline that preprocesses MRI scans, extracts meaningful features, trains two models, and compares accuracy on a held-out test set.

KEY RESULTS:

- Classical model accuracy: 70%*
- CNN (ResNet-18) accuracy: 81%*
- CNN detected 100% of tumor cases, making it the stronger diagnostic tool.*

SIGNIFICANCE:

Automated MRI tumor detection can support radiologists by speeding up screening and reducing subjectivity in diagnosis.



Introduction

Techniques Used:

- Otsu thresholding
- Morphological operations
- Connected components
- Feature extraction
- Logistic Regression
- CNN (ResNet-18 Transfer Learning)
- PyTorch & OpenCV

Dataset:

Brain MRI Images for Brain Tumor Detection (Kaggle)

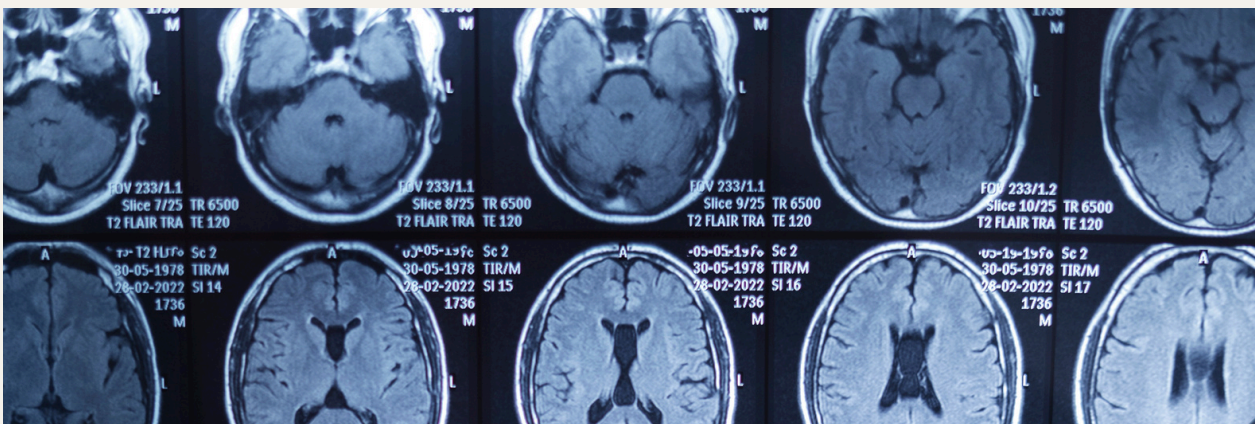


Image by Freepik — doctor looking at CT scan. Available at: https://www.freepik.com/free-photo/doctor-looking-ct-scan_25054001.htm (accessed 2025-11-25)

Motivation:

Early detection of brain tumors is critical for improving patient outcomes. My mom was diagnosed with stage 4 breast cancer at age 40, which influenced my interest in medical technology and inspired me to pursue this project.

Introduction Cont.



BACKGROUND:

Brain tumors are serious medical conditions that require early and accurate diagnosis. MRI scans are the preferred imaging method, but manual interpretation can be time-consuming and subjective.

PROBLEM:

Radiologists must review large volumes of MRI images, and subtle tumors may be difficult to identify without computational support.

Technical Challenge:

MRI tumors vary in shape, brightness, and position. Traditional handcrafted features may not generalize well. Deep models can learn richer representations but require more data.

This project compares classical and deep approaches to understand trade-offs.

Why Automation Helps:

Automated detection can:

- *highlight areas of concern*
- *reduce workload*
- *support second-opinion screening*
- *bring more consistency to diagnosis*

Methodology



Pipeline Overview:

The project consists of two main detection pipelines.

A. Classical Computer Vision Pipeline

- *Resize & denoise MRI*
- *Otsu thresholding*
- *Morphological opening & closing*
- *Connected components to isolate tumor region*
- *Compute features:*
 - *tumor area*
 - *bounding box width + height*
 - *aspect ratio*
 - *mean intensity*
- *Train Logistic Regression classifier*
- *Evaluate on a held-out test set*

B. Deep Learning Pipeline

(ResNet-18 Transfer Learning)

- *Convert grayscale → 3-channel*
- *Resize to 224×224*
- *Apply ImageNet normalization*
- *Freeze base layers*
- *Replace final FC layer with 2-class output*
- *Train for 5 epochs using Adam (lr= 1e-3)*
- *Choose best model by validation accuracy*
- *Evaluate on test set*

The project consists of two main detection pipelines. Although my proposal originally included handcrafted features before the CNN, I removed that step based on feedback. In the final project, the CNN is trained end-to-end directly on MRI pixels, following modern deep learning practices recommended by my instructor.

Experiments & Results



Classical Model Results (Logistic Regression)

Accuracy: 70%

Confusion Matrix
 $\begin{bmatrix} 7 & 4 \\ 4 & 12 \end{bmatrix}$

Interpretation:

Model performed reasonably well but lacked robustness due to limited feature complexity.

CNN Results (ResNet-18)

Accuracy: 81%

Tumor Recall: 1.00 (no missed tumors)

Confusion Matrix
 $\begin{bmatrix} 6 & 5 \\ 0 & 16 \end{bmatrix}$

Interpretation:

Deep learning learns richer texture and structural patterns, yielding higher accuracy and perfect tumor detection.

Failure Case Analysis:

Actual: no_tumor, Predicted: tumor

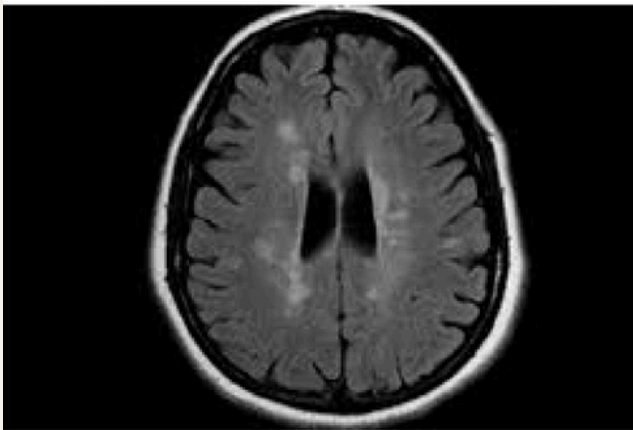


Figure 1. Example failure case from the CNN model. The model predicted "tumor" for a healthy MRI slice.

Although the CNN performed well overall, it still produced several false positives. One example is shown in Figure 1, where a healthy MRI slice (ground truth: no tumor) was mistakenly classified as tumor. In this case, the bright anatomical structures near the ventricles resemble the texture and contrast patterns of tumor regions seen in the training data. This suggests that the model sometimes relies on intensity-based cues rather than true pathological shapes. Including more diverse negative examples or applying better regularization could help reduce these errors.

Discussion & Challenges



Challenges:

- *Small dataset size*
- *Ensuring reliable segmentation masks*
- *Avoiding overfitting in CNN*
- *Class imbalance (more tumor cases)*
- *Complex environment setup on macOS Intel*

Strengths of Classical Approach:

- *Transparent/explainable*
- *Fast*
- *Good baseline*

Limitations:

- *Transparent/explainable*
- *Fast*
- *Cannot capture complex tumor shapes*
- *Lower accuracy*

Strengths of CNN:

- *Best performance*
- *Strong tumor sensitivity*
- *Learns meaningful hierarchical features*

Limitations:

- *More false positives*
- *Needs more data for specificity*

Conclusion & Future Work



Conclusion:

This project demonstrates that classical vision methods provide a solid baseline, but deep learning significantly improves accuracy in tumor classification. The ResNet-18 model achieved 81% accuracy with perfect tumor recall, highlighting its strong potential as a diagnostic support tool.

Future Work:

- *Fine-tune deeper CNN layers*
- *Add data augmentation*
- *Use class-weighted loss to reduce false positives*
- *Explore Grad-CAM heatmaps for interpretability*
- *Expand dataset for better generalization*